

Review

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Review

Beyond Dyadic Dialogue: A Comprehensive Survey of Multi-Party Dialogue Systems

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Abstract

Dialogue systems have evolved remarkably, from rule-based dyadic interfaces to large language model (LLM) powered conversational agents. Yet the predominant focus on two-party exchanges leaves a significant gap: most real-world communication, from business meetings to online group chats, is inherently **multi-party**. Multi-Party Dialogue (MPD) introduces new challenges in participant tracking, discourse structure modeling, and pragmatic reasoning that simpler dyadic systems are not equipped to handle. In this survey, we present a comprehensive examination of MPD research, organized along four axes: tasks, methods, datasets, and evaluation. We trace the methodological trajectory from early statistical and supervised models, through neural and graph-based approaches, to modern LLM-driven systems and emerging multi-agent paradigms. We further synthesize benchmark resources and evaluation protocols, and identify four central bottlenecks that constrain progress: combinatorial complexity of N-party dynamics, the structural-semantic divide, evaluation inadequacy, and data scarcity. We map these challenges to concrete research opportunities, offering a roadmap for building socially intelligent, holistically evaluated MPD systems.

Keywords: multi-party dialogue (MPD); large language models (LLMs); conversational agents; discourse structure modeling; evaluation protocols; multi-agent paradigms

1. Introduction

Dialogue systems form a foundational pillar of natural language processing (NLP). From early rule-based assistants to modern LLM-powered conversational agents, the field has primarily targeted dyadic interactions: a single user engaging with a single system. Typical applications include customer service, question-answering bots, task-oriented dialogue, and knowledge-grounded healthcare dialogue such as cognitive **stimulation** systems for elderly users [Jiang et al. \(2023\)](#). With a limited number of participants and relatively straightforward turn structure, these systems have benefited from modeling paradigms for tracking conversational state, contextual grounding, and response generation [Vaswani et al. \(2017\)](#).

However, dyadic settings capture only a slice of human communication. In meetings, classrooms, online forums, and collaborative platforms, conversations typically involve multiple participants who exchange information, negotiate intent, and share knowledge in parallel. Such **Multi-Party Dialogue (MPD)** is fundamentally more complex [Ishizaki and Kato \(1998\)](#); [Sapkota et al. \(2025\)](#). Three dimensions distinguish MPD from its dyadic counterpart. First, the participant dimension requires speaker tracking, addressee recognition, and turn management. Second, the discourse structure dimension involves frequent topic shifts, interleaved threads, and multi-level discourse organization. Third, the task dimension demands modeling of social roles, intent dynamics, and recurring interaction patterns alongside coherent generation.

Across these dimensions, MPD research has progressed through several methodological waves. Early systems relied on statistical and rule-based methods such as Hidden Markov Models, Conditional Random Fields, and finite-state machines for dialogue state tracking and intent recognition [Martínez-Hinarejos et al. \(2010\)](#); [Shang et al. \(2020\)](#); [Zhu et al. \(2010\)](#). The deep learning era introduced CNN, RNN, and LSTM architectures that improved nonlinear context modeling [Mangrulkar et al. \(2018\)](#); [Wen et al. \(2015\)](#); [Skantze \(2017\)](#). More recently, large language models such as GPT, T5, LLaMA, and Qwen have shown strong language understanding and generation capabilities, providing new affordances for long-context, multi-speaker reasoning. The latest frontier extends LLMs into multi-agent and hybrid paradigms, enabling autonomous role assignment, information sharing, and human-machine collaboration in group settings [Sapkota et al. \(2025\)](#); [Wang et al. \(2026\)](#).

Despite this progress, the field faces persistent obstacles. Existing models struggle to manage combinatorial information flow among speakers and to maintain coherent discourse structure across interleaved threads. Datasets remain limited in scale, modality, and annotation depth compared to dyadic resources. Evaluation protocols largely inherit metrics such as BLEU and F1 from NLP tasks, which fail to capture the social, structural, and collaborative qualities that matter in group conversation. Together, these gaps slow both methodological maturation and practical deployment.

This survey synthesizes the current state of MPD research with the goal of clarifying where the field stands and where it should go. Our contributions are as follows: (1) We propose a unified taxonomy of MPD tasks spanning six categories, from participant modeling to high-level decision support (Section 2). (2) We chart the methodological trajectory of MPD systems, organizing approaches from statistical models, through neural and graph-based methods, to modern LLM and multi-agent paradigms, with a complete catalog of 50+ methods (Section 3). (3) We catalogue representative datasets and evaluation metrics, highlighting their coverage, limitations, and complementarity (Sections 4 and 5). (4) We articulate four fundamental challenges facing MPD research and map each to concrete opportunities, providing a strategic roadmap for the next generation of socially intelligent dialogue systems (Section 6).

2. Tasks in Multi-Party Dialogue

MPD encompasses a diverse landscape of tasks that reflect its inherent multi-speaker, multi-thread complexity. Figure 1(a) provides a visual taxonomy. We organize existing tasks into six interdependent groups, presenting task definitions and representative work for each.

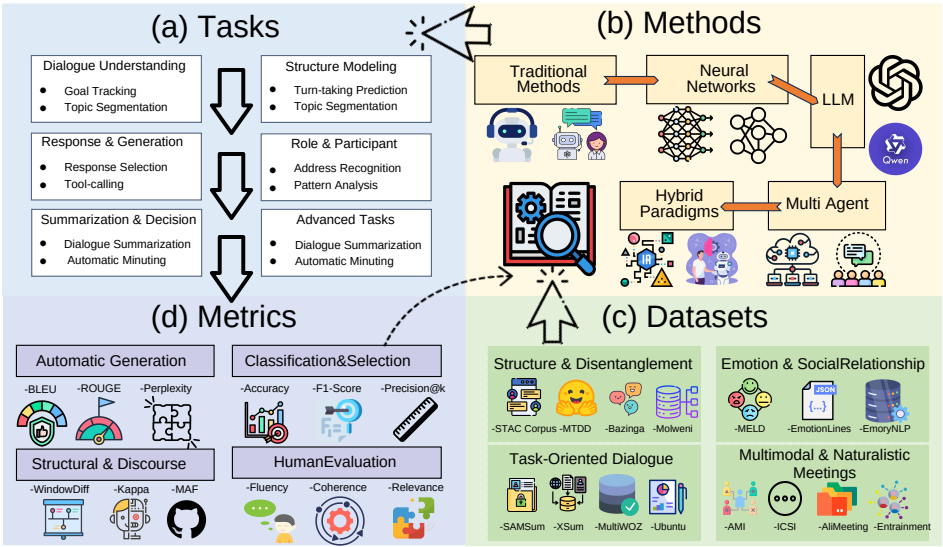


Figure 1. An overview of the Multi-Party Dialogue (MPD) research landscape, organized along four axes: (a) a six-category task taxonomy, (b) a methodological progression from traditional approaches through neural networks and large language models to hybrid multi-agent paradigms, (c) representative datasets grouped by purpose, and (d) four families of evaluation metrics. See Sections 2 to 5 for full details.

2.1. Dialogue Understanding

Dialogue understanding tasks aim to capture the semantic content of multi-party dialogues, such as goals, emotions, and speaker intentions. These tasks provide the foundation for higher-level modeling and reasoning.

Goal Tracking and Intent-Slot Recognition. This task involves detecting and tracking conversational goals as well as identifying semantic slots. In multi-party contexts, the difficulty arises from overlapping goals among participants and potential goal shifts during interactions [Addlesee et al. \(2023\)](#); [Penzo et al. \(2024\)](#). Recent work explores hybrid intent recognition frameworks that combine lightweight models such as BERT with LLMs under zero- and few-shot settings. By leveraging uncertainty-based routing and sharing probability-based information between models, these approaches improve intent recognition and out-of-scope detection performance in multi-party conversations while reducing computational costs [Castillo-López et al. \(2025\)](#).

Emotion Recognition in MPD. Recognizing emotions across multiple participants is crucial for building socially intelligent systems. Emotion dynamics in MPD are more complex than in dyadic dialogues, as emotional contagion and group-level sentiment often emerge [Sun et al. \(2021\)](#).

Dialogue Act Classification. Assigning functional roles (for example, question, answer, suggestion, agreement) to utterances supports downstream applications such as summarization and decision support. In MPD, dialogue act recognition must consider both local context and cross-speaker dependencies [Martinenghi et al. \(2024\)](#); [Mayfield et al. \(2012\)](#).

Multi-party Dialogue Reading Comprehension (MRC). MPD-MRC extends traditional machine reading comprehension to dialogue transcripts. It requires reasoning over multiple turns and speakers, integrating discourse and role information for accurate question answering [Li and Zhao \(2023, 2021\)](#).

Disruptive Talk Detection. Detecting disruptive or toxic utterances is important for maintaining constructive discussions in collaborative or educational settings [Park et al. \(2022\)](#).

Argument Prediction in Debates. In debate-style MPD, predicting which arguments are more persuasive or likely to win is a unique task with implications in computational social science [Sia et al. \(2022\)](#).

2.2. Dialogue Structure Modeling

These tasks aim to capture and represent the structural properties of MPD, which are often more complex than in dyadic dialogues.

Turn-taking Prediction and Modeling. MPD requires predicting when a participant will speak and determining the next speaker. This is challenging due to overlapping speech, interruptions, and implicit floor management strategies [Gatti de Bayser et al. \(2019\)](#); [Laskowski \(2010\)](#); [Enomoto et al. \(2020\)](#); [Wang et al. \(2024\)](#).

Topic Segmentation and Identification. Identifying topic boundaries and recognizing emerging topics in multi-party discussions is crucial for summarization and retrieval. Unlike single-topic dialogues, MPD often involves multiple threads of conversation [Purver et al. \(2006\)](#); [Galley et al. \(2003\)](#); [Nguyen et al. \(2013\)](#).

Dynamic Topic Disentanglement. Conversations in MPD are often interleaved. Disentanglement aims to separate intertwined threads into coherent topic-specific sub-dialogues [Wang et al. \(2020\)](#).

Discourse Parsing and Structure Prediction. Parsing discourse relations (for example, elaboration, contrast, agreement) and predicting conversation structures are essential for understanding the logical flow of MPD. Successful entry prediction, segmentation, and coherence modeling fall under this area [Mayfield et al. \(2012\)](#); [Afantenos et al. \(2015\)](#); [Li et al. \(2023\)](#); [Liu and Chen \(2021\)](#); [Liu et al. \(2025\)](#).

Decision-making Sub-dialogue Detection. Many MPD scenarios, such as meetings, involve decision-making. This task detects sub-dialogues where group decisions are negotiated [Fernández et al. \(2008\)](#); [Bui et al. \(2009\)](#).

Multi-turn Alignment and Chain-of-Thought Reasoning. Aligning utterances across turns and modeling explicit reasoning chains in MPD enable advanced reasoning-oriented applications [Kiruluta et al. \(2025\)](#).

2.3. Response and Generation

Response and generation tasks are at the heart of dialogue system development. In MPD, response generation is complicated by multiple potential addressees, shifting contexts, and social constraints.

Response Generation in MPC. Generating contextually appropriate responses in multi-party dialogues is one of the most fundamental challenges. It requires reasoning about speaker history, dialogue context, and social dynamics [Wang et al. \(2020\)](#); [Li et al. \(2023\)](#); [Fan et al. \(2024\)](#); [Gu et al. \(2022, 2023\)](#); [Hu et al. \(2025\)](#); [Liu et al. \(2019\)](#); [Tan et al. \(2023\)](#).

Response Selection. Given multiple candidate responses, the task is to select the most appropriate one. This often serves as an intermediate step for evaluating dialogue models [Penzo et al. \(2024\)](#); [Wang et al. \(2020\)](#); [Gu et al. \(2023, 2021a,b\)](#); [Ouchi and Tsuboi \(2016\)](#).

Response Generation with Addressee Prediction. Unlike dyadic dialogues, MPD responses must often include an explicit addressee, requiring models to jointly predict the response and its recipient [Li and Zhao \(2023\)](#).

Persona-aware Generation. This task focuses on maintaining speaker-specific personalities or roles across responses, which is especially relevant in long-term or role-based conversations [Ju et al. \(2022\)](#); [Mahajan and Shaikh \(2024\)](#).

Empathetic Dialogue Generation in MPD. Empathy is particularly complex in MPD, as systems must balance the emotions of multiple participants while maintaining coherence [Zhu et al. \(2022\)](#). This challenge is exemplified by group cognitive stimulation dialogue for elderly users with cognitive impairment, where an adaptive policy must coordinate therapeutic goals across several participants simultaneously [Jiang et al. \(2026\)](#).

Dialogue Generation with Next-Speaker Prediction. Extends response generation to also predict which participant should speak next, aligning generation with realistic conversation dynamics [Wang et al. \(2024\)](#).

Tool-calling in Multi-party Dialogue. Emerging work incorporates external APIs or tools (for example, calendars, databases) during multi-turn group conversations, making dialogue more interactive and actionable [Jang et al. \(2025\)](#). Beyond static tool invocation, recent work explores tool-driven intrinsic adaptation, where agents unify task execution with self-reconfiguration of their tool repertoire [Zhou et al. \(2026\)](#).

2.4. Role and Participant Modeling

Since MPD inherently involves multiple participants, modeling their roles, identities, and interactions is a crucial research focus.

Addressee Recognition and Identification. Identifying the intended recipient of an utterance is a fundamental task in MPD. It is closely tied to response generation and turn-taking modeling [Penzo et al. \(2024\)](#); [Gu et al. \(2023, 2021a,b\)](#); [Ouchi and Tsuboi \(2016\)](#); [Heylen and op den Akker \(2007\)](#); [Le et al. \(2019\)](#); [Zhu et al. \(2023\)](#).

Speaker Identification and Classification. Determining who is speaking in overlapping or noisy conditions is a prerequisite for many higher-level tasks [Gu et al. \(2023, 2021a,b\)](#); [Du et al. \(2022\)](#); [Meng et al. \(2018\)](#).

Overlapped Speaker Diarization. This task aims to separate overlapping speech and attribute it correctly to different participants, which is especially challenging in real-time settings [Du et al. \(2022\)](#).

Social Role and Seniority Inference. Inferring roles such as leader, moderator, or newcomer provides insights into group dynamics and social hierarchy [Laskowski et al. \(2008\)](#).

Interaction Pattern Analysis. Beyond individual roles, this task models recurring interaction structures, such as supportive behaviors, challenges, or questioning patterns [Heylen and op den Akker \(2007\)](#).

2.5. Summarization and Decision Support

Summarization and decision-support tasks seek to condense lengthy dialogues into concise, actionable outputs. These are particularly relevant in professional or collaborative contexts.

Dialogue Summarization. Generating abstractive or extractive summaries of group discussions. This requires modeling global context, topic transitions, and participant contributions [Chen and Yang \(2020\)](#).

Extractive Summarization in MPC. A variant of summarization that selects salient utterances instead of generating new text [Chen and Metze \(2012\)](#).

Automatic Minuting. Specialized summarization that produces structured meeting minutes with decisions, action items, and responsibilities [Bhatnagar et al. \(2022\)](#).

Decision Discussion Detection and Summarization. Identifying decision-related dialogues and producing concise summaries tailored to decision-making processes [Bui et al. \(2009\)](#).

2.6. Advanced and Emerging Tasks

Beyond classical tasks, researchers have explored advanced and domain-specific problems in MPD.

Machine Translation in MPD. Translating utterances in multilingual multi-party conversations while preserving role and context [Bhatnagar et al. \(2022\)](#).

Extractive Question Answering in MPC. Answering factual questions directly from conversation transcripts [Li et al. \(2023\)](#); [Zhou et al. \(2025\)](#).

Negotiation Strategy Learning. Learning strategies for successful negotiation in multi-party discussions, applicable to economics and multi-agent systems [Hiraoka et al. \(2015\)](#).

Entity Linking in MPC. Linking entity mentions within dialogues to external knowledge bases, facilitating knowledge-grounded conversations [Aina et al. \(2019\)](#).

Crucially, these tasks are deeply *interdependent*. Accurate addressee recognition strengthens response generation, while reliable discourse parsing improves summarization. This interdependence motivates the recent trend toward holistic, multi-task modeling discussed in Section 6.

3. Methods in Multi-Party Dialogue

The methodological history of MPD mirrors the broader arc of NLP, progressing from statistical models, through neural and graph-based architectures, to large language models and multi-agent systems. Figure 2 visualizes this evolution along four major research branches that have continuously matured over two decades, while Table 1 highlights representative methodological milestones along the trajectory. The complete catalog of 50+ methods appears in Table 4, grouped chronologically into the statistical/early-neural era (pre-2020), the pre-trained model and graph era (2020–2023), and the LLM and multi-agent era (2024–2026).

Table 1. Representative milestones in MPD research, illustrating the field’s evolution from statistical models, through neural and graph-based architectures, to LLMs and multi-agent paradigms. Headline results are reported on each paper’s main benchmark. The complete catalog appears in Table 4.

Method	Task	Dataset	Era	Headline Result
Feature-driven seg. Galley et al. (2003)	Topic segmentation	ICSI	Statistical	$P_k=23.0$, $WD=25.47$
Bayesian topic models Purver et al. (2006)	Topic segmentation	ICSI	Statistical	$P_k=28.9$, $WD=32.9$
Static/Dynamic RNN Ouchi and Tsuboi (2016)	Addressee + Response	Self-built	Neural	$ADR=68.54$, $RES=78.64$
W2W Le et al. (2019)	Addressee identification	Ubuntu IRC	Neural	$Len=5=80.86$
MPC-BERT Gu et al. (2021b)	Addressee, Speaker, Sel.	Ubuntu IRC	PLM	$P@1=98.31$, $Acc=92.42$
HeterMPC Gu et al. (2022)	Response generation	Ubuntu IRC	Graph + PLM	$BLEU-1=12.61$
GIFT Gu et al. (2023)	Multi-task understanding	Ubuntu IRC	Graph + PLM	$R_2@1=95.04$
ELECTRA-EMVI Li et al. (2023)	Discourse parsing, QA	Molweni	PLM + Latent	$F1_G=91.78$
PFT-Prompt Addlesee et al. (2023)	Goal tracking	EU SPRING	LLM Prompting	$Acc=69.57$
RL-TRC Fan et al. (2024)	Response generation	Ubuntu IRC	RL + LLM	$BLEU-1=13.66$
MuPaS Wang et al. (2024)	Gen. + Speaker pred.	Friends, GoT	LLM-SFT	$GSM8K=43.14$
DICE-BENCH Jang et al. (2025)	Tool-calling	DICE-BENCH	Multi-Agent	$DICE=3.64$
SS-MPC Jang et al. (2025)	Response generation	Ubuntu IRC	Multi-Agent	$BLEU-1=15.60$
EverMemBench Hu et al. (2026)	Long-horizon memory	EverMemBench	Multi-Agent	$Avg=37.44/72.61$

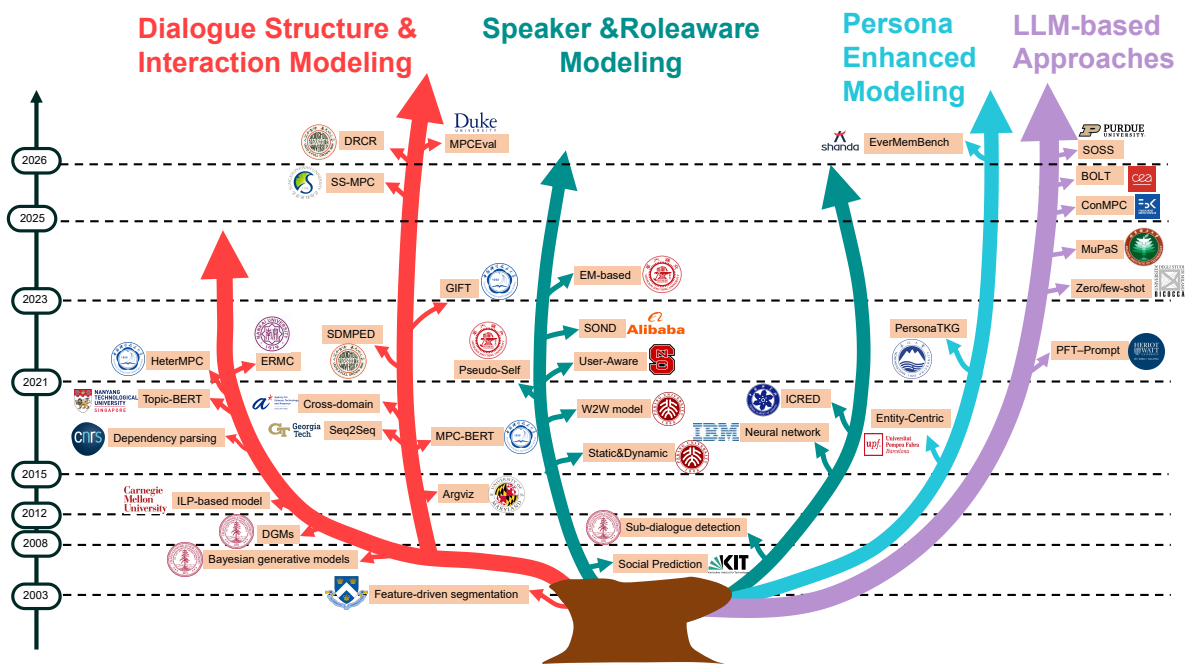


Figure 2. Historical evolution of representative MPD methods from 2003 to 2026, organized along four research branches: *Dialogue Structure and Interaction Modeling* (red), *Speaker and Role-aware Modeling* (teal), *Persona Enhanced Modeling* (cyan), and *LLM-based Approaches* (purple). The tree visualization highlights how each branch matures from early statistical models to graph neural networks and modern LLM-driven systems. See Section 3 for detailed discussion.

3.1. From Statistical Models to Neural Networks

Early MPD systems leveraged statistical learning. Classical models for next-speaker prediction (MLE, SVM, CNN) demonstrated the value of semantic features in turn dynamics Gatti de Bayser et al. (2019). Beyond turn dynamics, statistical approaches addressed a wide range of structural and social tasks. Unsupervised Bayesian generative topic models offered robustness to ASR noise and domain shift for topic segmentation in noisy meeting transcripts Purver et al. (2006), while feature-driven segmentation combined lexical and acoustic cues for domain-agnostic boundary detection Galley et al. (2003). ILP formulations with sociolinguistic constraints tackled hierarchical structure prediction Mayfield et al. (2012), MST-based dependency parsing modeled discourse structure in multi-party chat Afantenos et al. (2015), and structured labeling together with directed-graph methods detected and summarized decision-related sub-dialogues Fernández et al. (2008); Bui et al. (2009). Complementary lines of work predicted social dimensions and seniority from interaction patterns Laskowski et al. (2008), analyzed backchannel and gaze cues for interaction modeling Heylen and op den Akker (2007), applied random-walk extractive summarization to meeting transcripts Chen and Metze (2012), learned negotiation strategies in simulated trading via reinforcement learning Hiraoka et al. (2015), and linked entity mentions to knowledge bases Aina et al. (2019).

The shift to neural architectures introduced richer context modeling. Joint static-dynamic RNN models tackled addressee and response selection together Ouchi and Tsuboi (2016), W2W learned to identify utterance addressees from raw context Le et al. (2019), and neural speaker-classification models compared temporal, content-driven, and hybrid representations, revealing the benefit of integrating sequential and semantic information Meng et al. (2018). Context encodings that integrate speaker and addressee role information improved response generation Liu et al. (2019), and BERT-based multi-task models enabled dynamic topic tracking for reply selection Wang et al. (2020). These neural advances laid the groundwork for the pre-trained model era that followed.

3.2. Pre-trained Models and Graph Reasoning

Pre-trained language models reshaped MPD research. MPC-BERT jointly models speaker identity and utterance semantics through self-supervised tasks including reply-to recognition, identical speaker searching, and pointer consistency distinction [Gu et al. \(2021a,b\)](#), while latent discourse inference exploits unlabeled multi-party data efficiently [Li et al. \(2023\)](#). Pseudo-self-supervised pre-training likewise improves multi-party reading comprehension and dialogue QA [Li and Zhao \(2021\)](#), and cross-domain Transformer parsers transfer discourse-parsing capability across corpora [Liu and Chen \(2021\)](#). To address the graph-structured nature of MPD, heterogeneous graph neural networks such as HeterMPC [Gu et al. \(2022\)](#) and MADNet [Gu et al. \(2023\)](#) explicitly model utterances and interlocutors as typed nodes with multiple meta-relations, with EM-based optimization handling missing addressee data. Lightweight graph-induced fine-tuning (GIFT) integrates structural signals into pre-trained Transformers with only a handful of additional parameters per encoding layer [Gu et al. \(2023\)](#), and persona-aware variants enhance speaker consistency through personality-based embeddings [Ju et al. \(2022\)](#); [Mahajan and Shaikh \(2024\)](#). Global-local graph reasoning further supports multi-hop question answering over complex multi-party contexts [Li and Zhao \(2023\)](#), and graph-aware sentiment models capture emotion propagation across speakers [Sun et al. \(2021\)](#). The same period also saw multi-view BART for abstractive summarization [Chen and Yang \(2020\)](#), static-dynamic empathy frameworks (SDMPED) for emotion-aware generation [Zhu et al. \(2022\)](#), discrete addressee codebook encodings for robust addressee recognition [Zhu et al. \(2023\)](#), and neural diarization models such as SOND for overlapped speech [Du et al. \(2022\)](#).

3.3. Large Language Models and Multi-Agent Paradigms

Recent work probes the multi-party capabilities of large language models. Systematic evaluations expose where ChatGPT and GPT-4 succeed and fail in multi-party reasoning [Penzo et al. \(2024\)](#); [Tan et al. \(2023\)](#), and zero- and few-shot LLMs are tested on dialogue act classification [Martinenghi et al. \(2024\)](#). Reasoning-oriented prompting often outperforms conventional fine-tuning for goal tracking and intent-slot recognition [Addlesee et al. \(2023\)](#), and hybrid intent recognition frameworks combine BERT-scale models with LLMs through uncertainty routing to balance accuracy and cost [Castillo-López et al. \(2025\)](#). Reinforcement learning approaches optimize response coherence and negotiation outcomes, using utterance-level rewards [Fan et al. \(2024\)](#); [Hiraoka et al. \(2015\)](#) and cross-attention historical reward functions for self-supervised, context-aware generation [Kiruluta et al. \(2025\)](#); reward-decomposition strategies that mine latent logic further enable self-evolving reasoning [Chen et al. \(2026\)](#); while logical reasoning memory networks enhance QA accuracy [Zhou et al. \(2025\)](#). LLM-based explainable discourse parsing integrates contrastive learning to improve utterance-level accuracy [Liu et al. \(2025\)](#), and contrastive history-aware models further refine response consistency [Hu et al. \(2025\)](#).

The latest frontier extends MPD into multi-agent territory. MuPaS explores supervised fine-tuning for group dialogue generation with next-speaker prediction [Wang et al. \(2024\)](#); DICE-BENCH evaluates tool-calling in multi-round, multi-party scenarios with 1,607 synthesized dialogues [Jang et al. \(2025\)](#); SS-MPC encodes dialogue structure directly as sequential inputs within a Transformer encoder-decoder framework, removing the need for graph representations [Jang et al. \(2025\)](#); discourse-coherence response-guided context models (DRCR) improve generation on Ubuntu IRC [Cao et al. \(2026\)](#); context-aware turn-taking models predict when an agent should speak or stay silent [Bhagtani et al. \(2026\)](#); and MPCEval introduces a multi-dimensional benchmark for multi-party conversation quality [Zhang et al. \(2026\)](#). EverMemBench probes long-horizon collaborative memory across multi-group settings, revealing severe limitations in multi-hop attribution reasoning and temporal revision tracking by current LLM-based systems [Hu et al. \(2026\)](#). Synthetic data generation also emerges as a scalable resource for downstream MPD tasks: deterministic-constraint frameworks [Penzo et al. \(2026\)](#), tree-guided subspace partitioning for diverse data synthesis [Wang et al. \(2026\)](#), and knowledge-driven progressive-thought prompting for multi-turn dialogue augmentation [Jiang et al. \(2024\)](#) all help mitigate data scarcity. Orthogonally, parameter-efficient fine-tuning methods such as low-rank

adaptation and its rotation- and shard-based variants make it practical to specialize large models for multi-party applications under limited compute Wang et al. (2024a,b, 2025).

Takeaway. The trajectory in Figure 2 reveals a clear pattern: as model capacity grows, attention shifts from explicit structural modeling toward generative reasoning. Yet graph-based and symbolic approaches retain their value for capturing the relational structure that pure sequential models miss. Increasingly, state-of-the-art systems combine multiple paradigms, hinting at the neuro-symbolic hybrid architectures we identify as a key opportunity in Section 6.

4. Datasets

The diversity of MPD tasks is mirrored in the breadth of available corpora, which differ in source, scale, modality, and annotation depth. Table 2 summarizes representative datasets by category, providing a comparison of their scale and annotation focus. We organize them into four groups.

Table 2. Representative MPD datasets organized by category, with key statistics and annotation focus. Categories are: **Struct.** (Structure and Disentanglement), **Emo.** (Emotion and Social Relationship), **Task.** (Task-Oriented and Open-Domain), and **Multi.** (Multimodal and Naturalistic Meetings).

Dataset	Category	Source	Scale	Modality	Key Annotations
STAC Asher et al. (2016)	Struct.	Online games	Multi-party chats	Text	SDRT discourse structure, dialogue acts
Molweni Li et al.	Struct.	Ubuntu chat	10K dial./88K utt.	Text	Discourse dependency, QA pairs
MTDD Chang et al. (2023)	Struct.	TV/movie scripts	10K turns/831 shows	Text	Thread disentanglement, floor changes
Bazinga! Lerner et al. (2022)	Struct.	TV/movie scripts	Large-scale	Text	Diarization, addressee, entity linking
MELD Poria et al. (2019)	Emo.	Friends	13K utterances	A+V+T	Emotion, sentiment (multimodal)
EmotionLines Hsu et al. (2018)	Emo.	Friends/FB	29K utterances	Text	Seven emotion labels
EmoryNLP Zahiri and Choi (2017)	Emo.	Friends	12.6K utterances	Text	Seven crowdsourced emotions
MPDD Chen et al. (2020)	Emo.	Chinese TV	Multi-party	Text	Emotions, interpersonal relations (CN)
ALOHA Li et al. (2020)	Emo.	TV/movie scripts	1M+ dialogue lines	Text	Human-Level Attributes (HLAs)
SAMSum Gliwa et al. (2019)	Task.	Messenger style	16K conversations	Text	Third-person abstractive summaries
XSum Narayan et al. (2018)	Task.	BBC articles	226K pairs	Text	Single-sentence extreme summaries
MultiWOZ Budzianowski et al. (2020)	Task.	Wizard-of-Oz	7 domains	Text	Belief tracking, dialogue acts
Ubuntu Lowe et al. (2015)	Task.	Tech support IRC	~1M dialogues	Text	Next-response selection
DICE-BENCH Jang et al. (2025)	Task.	Synthesized	1,607 dialogues	Text	Multi-round tool-calling traces
WMPC Penzo et al. (2026)	Task.	LLM-synthesized	Constrained	Text	Structure, stance, interaction flow
AMI Carletta et al. (2006)	Multi.	Real meetings	100 hours	A+V+T	Dialogue acts, gestures, attention
ICSI MRDA Shriberg et al. (2004)	Multi.	Real meetings	72 h / 180K DA	A+T	Dialogue acts, adjacency pairs
AliMeeting Yu et al. (2022)	Multi.	Mandarin meetings	120 hours	Audio (8ch)	Speaker diarization, multi-speaker ASR
Teams Corpus Litman et al. (2016)	Multi.	Cooperative games	63 teams	A+V+T	Entrainment, dominance, group dynamics

4.1. Structure and Disentanglement

Datasets in this category support research on threads, discourse relations, and conversational structure. The **STAC Corpus** Asher et al. (2016), released at LREC 2016, contains multi-party chat dialogues annotated with full discourse structure based on Segmented Discourse Representation Theory (SDRT), including both linguistic and extra-linguistic interactions, enabling research on dialogue act classification and discourse-level reasoning. **Molweni Li et al.**, presented at COLING 2020, is an MRC dataset derived from the Ubuntu Chat Corpus consisting of 10,000 dialogues and 88,303 utterances annotated with discourse dependency structures based on a modified SDRT. **MTDD** Chang et al. (2023) comprises 10,033 turns from 831 movies and TV series, annotated for conversational thread disentanglement and floor changes. **Bazinga!** Lerner et al. (2022) consists of large-scale multi-party dialogues from TV and movie scripts with rich annotations for speaker diarization, addressee recognition, and entity linking, while **OnlineChat-MPCorpus** Shaikh et al. (2010) focuses on online chatroom

conversations annotated at four levels: conversational links, dialogue acts, local topics, and meso-level discourse structure.

4.2. Emotion and Social Relationship

TV transcripts dominate this category. **MELD** Poria et al. (2019), released at ACL 2019, contains approximately 13,000 utterances from *Friends*, providing audio, visual, and textual modalities annotated with fine-grained emotion and sentiment labels. Its textual precursor **EmotionLines** Hsu et al. (2018) includes 29,245 utterances from *Friends* and private Facebook chats annotated with seven emotion labels, and **EmoryNLP** Zahiri and Choi (2017) utilizes *Friends* transcripts (S1–S4), comprising 12,606 utterances with seven crowdsourced labels. **MPDD** Chen et al. (2020) supports Chinese-language affective MPD with annotations for emotions and interpersonal relationships, and **ALOHA (HLA-Chat)** Li et al. (2020) links over a million dialogue lines from TV shows and movies to “Human-Level Attributes” (HLAs) based on audience-determined tropes for persona modeling.

4.3. Task-Oriented and Open-Domain

SAMSum Gliwa et al. (2019) contains 16,369 messenger-like conversations with manually written third-person summaries, a primary benchmark for abstractive dialogue summarization, while **XSum** Narayan et al. (2018) focuses on extreme summarization with 226,711 BBC article-summary pairs. **MultiWOZ** Budzianowski et al. (2020) is a large-scale human-human dataset for task-oriented dialogue spanning seven domains, and the **Ubuntu Dialogue Corpus** Lowe et al. (2015) contains nearly one million dialogues from technical support IRC logs for next-response selection. **IncTempSum** Manuvinakurike et al. (2021) focuses on incremental abstractive summarization combining crowdsourced and expert annotations. More recently, **DICE-BENCH** Jang et al. (2025) introduces 1,607 synthesized multi-round, multi-party dialogues for tool-use evaluation, and synthetic **WMPC** generation frameworks Penzo et al. (2026) demonstrate that instruction-tuned LLMs can produce structurally diverse MPD data under deterministic constraints such as dialogue structure and speaker stance, with turn-by-turn generation improving constraint adherence and linguistic variability. **ChatEval** Sedoc et al. (2019) is designed for the evaluation of open-domain chatbots with manually filtered utterances and reference responses.

4.4. Multimodal and Naturalistic Meetings

The **AMI Meeting Corpus** Carletta et al. (2006) offers 100 hours of meeting recordings (natural and elicited) with multimodal annotations including dialogue acts, head/hand gestures, and focus of attention. The **ICSI Meeting Corpus (MRDA)** Shriberg et al. (2004) supplies 72 hours of naturally occurring professional meetings featuring over 180,000 hand-annotated dialogue-act tags. **AliMeeting (M2MeT)** Yu et al. (2022) provides 120 hours of real-world Mandarin meetings recorded with 8-channel microphone arrays for diarization and multi-speaker ASR, while the **Teams Corpus** Litman et al. (2016) captures realistic cooperative group dynamics across 63 teams through aligned audio, video, and detailed transcripts annotated for entrainment and dominance.

Despite this variety, current corpora remain skewed toward text-only English data, leaving multimodal, multilingual, and naturalistic professional settings comparatively underexplored, a gap that remains critical and we revisit in Section 6.

5. Evaluation

Evaluating MPD systems is harder than dyadic ones because quality spans multiple dimensions: semantic accuracy, structural coherence, social appropriateness, and collaborative effectiveness. Table 3 groups four metric families adopted in MPD studies, with representative metrics and typical use cases.

Table 3. The four families of MPD evaluation metrics, with representative metrics and primary use cases.

Family	Representative Metrics	Primary Use Cases
Automatic Generation	BLEU, ROUGE, METEOR, BERTScore, PPL, Distinct-n	Response generation, summarization, fluency, diversity
Classification & Selection	Accuracy, F1 (macro/micro), Precision@k, Recall@k, MRR, EM, AUC, DICE-score	Addressee/intent classification, response selection, tool-calling
Structural & Discourse	P_k , WindowDiff, Link/Link+Rel F1, DER, Kappa	Topic segmentation, discourse parsing, diarization
Human Evaluation	Fluency, Coherence, Relevance, Informativeness, Authority	Pragmatic appropriateness, role coordination

5.1. Automatic Generation Metrics

N-gram overlap measures including BLEU (BLEU-1 through BLEU-4) [Li and Zhao \(2023\)](#); [Fan et al. \(2024\)](#); [Zhu et al. \(2022\)](#); [Bhatnagar et al. \(2022\)](#), SacreBLEU [Tan et al. \(2023\)](#), and ROUGE (ROUGE-1, ROUGE-L) [Li and Zhao \(2023\)](#); [Zhu et al. \(2022\)](#); [Chen and Yang \(2020\)](#); [Chen and Metze \(2012\)](#); [Bhatnagar et al. \(2022\)](#) remain widely used for summarization and response generation. BLEU measures the precision of n-gram matches, while ROUGE emphasizes recall and is especially suitable for summarization. METEOR adds stemming, synonym matching, and word-order sensitivity [Li and Zhao \(2023\)](#); [Afantenos et al. \(2015\)](#); [Fan et al. \(2024\)](#); [Gu et al. \(2022\)](#). For semantic similarity, BERTScore computes overlap using contextual embeddings [Ju et al. \(2022\)](#); [Bhatnagar et al. \(2022\)](#), and other embedding-based metrics reduce reliance on lexical overlap [Ju et al. \(2022\)](#). Perplexity (PPL) gauges fluency by measuring the likelihood assigned to human-written references [Laskowski \(2010\)](#); [Ju et al. \(2022\)](#), though lower PPL does not guarantee relevance. Finally, Distinct-n (for example, Dist-1/2) quantifies diversity by computing the ratio of unique n-grams [Ju et al. \(2022\)](#), which is particularly relevant in MPD where repetitive answers harm conversational quality.

5.2. Classification and Selection Metrics

Accuracy, precision, recall, and F1 dominate addressee and intent classification, with variants including in-domain versus cross-domain accuracy [Addlesee et al. \(2023\)](#); [Sun et al. \(2021\)](#); [Mayfield et al. \(2012\)](#); [Gatti de Bayser et al. \(2019\)](#); [Gu et al. \(2021a,b\)](#); [Le et al. \(2019\)](#), and macro-, micro-, and weighted-F1 providing nuanced evaluation under class imbalance [Wang et al. \(2020\)](#); [Afantenos et al. \(2015\)](#); [Li et al. \(2023\)](#); [Liu et al. \(2025\)](#); [Gu et al. \(2021b\)](#); [Meng et al. \(2018\)](#); [Aina et al. \(2019\)](#). Ranking metrics such as Precision@k [Gu et al. \(2023, 2021a,b\)](#); [Le et al. \(2019\)](#), Recall@k [Gu et al. \(2023, 2021b\)](#), and Mean Reciprocal Rank [Meng et al. \(2018\)](#) evaluate candidate selection in retrieval-style tasks. Task-specific scores include ADR/RES for addressee-response pairs [Ouchi and Tsuboi \(2016\)](#), Exact Match (EM) for slot filling and structure prediction [Li and Zhao \(2023, 2021\)](#); [Li et al. \(2023\)](#); [Zhou et al. \(2025\)](#), AUC and PR-AUC for probabilistic classification [Park et al. \(2022\)](#); [Sia et al. \(2022\)](#); [Li et al. \(2023\)](#), and the recently introduced DICE-score for structured tool-calling [Jang et al. \(2025\)](#).

5.3. Structural and Discourse Metrics

Topic segmentation is assessed with P_k and WindowDiff (WD) [Purver et al. \(2006\)](#); [Galley et al. \(2003\)](#), with lenient-match variants adapting them to noisy data [Fernández et al. \(2008\)](#). Discourse parsing employs graph-prediction F1 and relation-classification F1 [Li et al. \(2023\)](#) together with Link and Link&Rel metrics [Liu et al. \(2025\)](#) to assess discourse connectivity. Diarization Error Rate (DER) measures speaker attribution accuracy [Du et al. \(2022\)](#). Some works adopt multi-level evaluation, distinguishing sentence-, sequence-, and thread-level scores using Accuracy, Kappa agreement (κ), authoritativeness r^2 , and Micro-averaged F-score [Mayfield et al. \(2012\)](#).

5.4. Human Evaluation

Despite advances in automatic scoring, human judgment remains indispensable for assessing fluency (grammatical correctness and naturalness) [Fan et al. \(2024\)](#); [Liu et al. \(2019\)](#); [Ju et al. \(2022\)](#); [Gu et al. \(2023\)](#), consistency and coherence (logical flow within context) [Kiruluta et al. \(2025\)](#); [Liu et al. \(2019\)](#); [Ju et al. \(2022\)](#); [Gu et al. \(2023\)](#), relevance (appropriateness to dialogue state) [Fan et al. \(2024\)](#); [Liu et al. \(2019\)](#); [Gu et al. \(2023\)](#), informativeness (usefulness of content) [Fan et al. \(2024\)](#); [Liu et al. \(2019\)](#); [Gu et al. \(2023\)](#), authority and trustworthiness (credibility) [Mayfield et al. \(2012\)](#), and interactional aspects such as step correctness, latency, and coherence [Kiruluta et al. \(2025\)](#). Task-specific measures further enrich the toolkit, including average reward in RL settings [Hiraoka et al. \(2015\)](#), cross-domain performance gap as a robustness measure [Liu and Chen \(2021\)](#), error reduction rate [Laskowski et al. \(2008\)](#), and composite scores that combine PPL, BLEU, Distinct, embedding similarity, accuracy, P/R/F1, and human ratings [Wang et al. \(2024\)](#); [Ju et al. \(2022\)](#).

Taken together, current evaluation practice remains fragmented: most studies rely on a handful of automatic metrics that imperfectly capture the social and collaborative dimensions of MPD. Section 6 argues for holistic, interaction-oriented benchmarks as a primary research opportunity.

6. Challenges and Opportunities

Building effective MPD systems requires confronting challenges distinct from dyadic dialogue. Figure 3 maps four bottlenecks to opportunities. We discuss each pair.

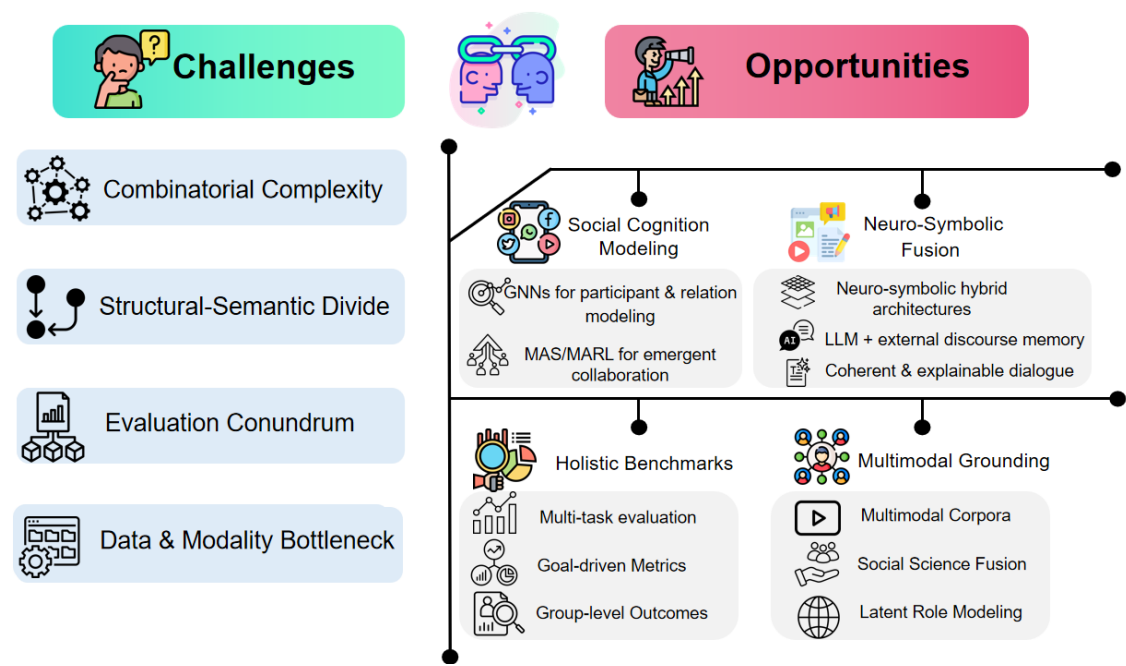


Figure 3. Strategic Roadmap of Challenges and Opportunities in Multi-Party Dialogue. The diagram identifies four fundamental bottlenecks hindering current MPD systems (Combinatorial Complexity of N-party dynamics, the Structural-Semantic Divide in discourse modeling, the Evaluation Conundrum, and the Data and Modality Bottleneck) and maps them to promising future research directions (Social Cognition Modeling, Neuro-Symbolic Fusion, Holistic Benchmarks, and Multimodal Grounding). See Section 6.

6.1. Combinatorial Complexity vs. Social Cognition

The transition from one-to-one to many-to-many interaction introduces a combinatorial explosion in conversational state. Models must track not only utterance history but also a dynamic graph of speakers, addressees, floor holders, and sub-threads. Current sequential LLMs flatten this structure, leading to failures in addressee prediction and contextually grounded turn-taking. The opportunity lies in architectures with inherent *social cognition*. Heterogeneous Graph Neural Networks naturally encode participants, utterances, and their evolving relationships as a dynamic graph. Multi-Agent Systems and

Reinforcement Learning enable autonomous agents to learn emergent communication strategies through simulated interaction, moving beyond pattern imitation toward goal-driven social behavior.

6.2. *The Structural-Semantic Divide vs. Neuro-Symbolic Fusion*

In MPD, meaning is entangled with discourse position. Conversations comprise interleaved threads where a speaker may respond to an utterance several turns prior. While LLMs capture local coherence, they lack explicit mechanisms for global hierarchical structure, producing fluent but logically fragmented contributions. Neuro-symbolic hybrid architectures offer a promising path: an LLM augmented with a queryable discourse graph that tracks threads, stances, and decision states. This fusion would enable responses that are fluent and globally coherent, with explainable reasoning over conversational history.

6.3. *The Evaluation Conundrum vs. Holistic Benchmarks*

The field leans heavily on metrics borrowed from other NLP tasks, which are poor proxies for conversational quality in group settings. BLEU and F1 fail to capture pragmatic appropriateness, role fulfillment, consensus building, and group-level outcomes such as decision quality. The opportunity is to design *holistic, multi-dimensional benchmarks* that evaluate interdependent capabilities jointly (for example, response generation together with addressee prediction and topic tracking) alongside *goal-oriented and human-in-the-loop protocols* that measure functional contribution to group outcomes rather than surface mimicry. Progress in adjacent areas is instructive: trajectory-aware evaluation of research agents [Chen et al. \(2026\)](#) and large-scale benchmarks for dialogue understanding and generation [Hu et al. \(2026\)](#) show how multi-dimensional, process-level evaluation can move beyond single-score metrics.

6.4. *The Data and Modality Bottleneck vs. Multimodal Grounding*

Progress is constrained by the scarcity of large-scale, richly annotated MPD corpora, a problem compounded for non-English settings where dedicated benchmarks and datasets remain limited, as recent work on Cantonese capabilities and resources illustrates [Jiang et al. \(2025a,b\)](#). Most existing data is text-only, missing the gaze, gesture, prosody, and turn-yielding cues that humans use to manage group conversation. The path forward involves large-scale multimodal corpora and, more profoundly, deeper integration with computational social science methods. By learning to infer latent social variables such as leadership, facilitation, and group cohesion from conversational data, AI can evolve from a passive participant into an active, socially aware collaborator capable of adapting to pragmatic and cultural context. Recent advances in multi-modal proactive reasoning, which decouple perception from reasoning, point toward systems that can ground group interaction in visual as well as textual signals [Zhou et al. \(2025\)](#).

7. Conclusion

Surveyed Multi-Party Dialogue research across tasks, methods, datasets, and evaluation. The field advanced significantly, from statistical models through deep learning and graph reasoning to large language models and multi-agent paradigms. Yet gaps remain: approaches struggle with N-party interaction complexity, evaluation protocols inadequately capture social and collaborative quality, and data resources lag behind dyadic dialogue. The path forward calls for integrated modeling frameworks jointly addressing multiple MPD tasks, deeper grounding in external knowledge and social dynamics, principled integration of multi-agent reasoning with LLM capabilities, human-centered evaluation, and richer multimodal datasets covering diverse collaborative scenarios. Realizing these directions requires interdisciplinary collaboration spanning linguistics, sociology, psychology, and artificial intelligence. We hope this survey provides a clear synthesis of current progress and a practical roadmap for socially intelligent dialogue systems capable of supporting human-human and human-machine collaboration in complex environments.

Limitations

This survey synthesizes MPD research across tasks, methods, datasets, and evaluation, but has limitations. It may miss recent or domain-specific studies, especially on LLMs, multi-agent systems, and synthetic data. The taxonomy is analytical rather than disjoint, and reported results are not directly comparable. We summarize metrics without new meta-evaluation and focus mainly on NLP-oriented MPD work. These limitations motivate unified and interdisciplinary future research.

Table 4. Comprehensive catalog of 50+ MPD methods, grouped chronologically into the Statistical and Early Neural Era (pre-2020), the Pre-trained Model and Graph Era (2020–2023), and the LLM and Multi-Agent Era (2024–2026). The table spans multiple pages.

Method	Domain/Task	Datasets	Metrics	Headline Result	Code
Statistical and Early Neural Era (pre-2020)					
Feature-driven Seg. Galley et al. (2003)	Topic seg.	ICSI	P_k ; WD	P_k =23.0	LDC
Bayesian Models Purver et al. (2006)	Topic seg.	ICSI	P_k ; WD	P_k =28.9	None
Backchannel Heylen and op den Akker (2007)	Interaction analysis	AMI	κ	κ =0.14	None
Sub-dialogue Det. Fernández et al. (2008)	Decision detect.	AMI	P; R; F1	F1=34	Stanford
Social Dim. Pred. Laskowski et al. (2008)	Social structure	AMI; ICSI	RERR	37 to 67%	None
Directed Graphs Bui et al. (2009)	Decision detect.	AMI	P; R; F1	F1=0.55	BNT
ILP Disentangle Mayfield et al. (2012)	Structure pred.	Cancer Support	MAF; κ	Acc=0.78; κ =0.60	None
Random Walk Chen and Metzger (2012)	Meeting sum.	SmartNotes	ROUGE	R-1=49.79	None
Argviz Nguyen et al. (2013)	Topic analysis	CNN Crossfire	Qualitative	Qualitative	Google toolkit
Dep. Parser Afantenos et al. (2015)	Structure parsing	STAC	F1	68.0 (unlabeled)	None
RL Trading Hiraoka et al. (2015)	Negotiation	Sim. trading	Avg. reward	Better than baseline	None
Static+Dynamic Ouchi and Tsuboi (2016)	ADR + RES sel.	Self-built	ADR; RES	ADR=68.54; RES=78.64	None
Neural Speaker Meng et al. (2018)	Speaker class.	Self-built	F1; MRR	Macro F1=44.25	Google Sites
W2W Le et al. (2019)	Addressee ID	Ubuntu IRC	P@n; Len-n	Len-5=80.86	None
ICRED Liu et al. (2019)	Response gen.	RGMPC	BLEU; ROUGE	Length=11.34	fasttext
ML Models Gatti de Bayser et al. (2019)	Turn-taking pred.	MultiWoZ; Finch	Accuracy	Acc=86.38	gensim
Entity-centric Aina et al. (2019)	Entity linking	Self-built	F1; Accuracy	F1=52.5; Acc=77.6	GitHub
Pre-trained Model and Graph Era (2020–2023)					
Multi-view Chen and Yang (2020)	BART Summarization	SAMSum	ROUGE	R-1=0.493	GT-SALT (GitHub)
Topic-BERT Wang et al. (2020)	Selection; topics	Ubuntu	Recall; MRR	R@10=97.0	salesforce (GitHub)
Turn-taking Annot. Enomoto et al. (2020)	Turn analysis	Chiba	POS; prosody	Qualitative	None
Pseudo-SSL Li and Zhao (2021)	Dialogue QA	FriendsQA; Molweni	EM; F1	EM=58.0; F1=72.9	EricLee8 (GitHub)
Cross-domain Trans. Liu and Chen (2021)	Discourse parsing	STAC; Molweni	F1 (link); L+R	Link=80.2	HuggingFace
MPC-BERT Gu et al. (2021b)	Addressee; speaker; selection	Ubuntu IRC	P@1; Acc	P@1=98.31; Acc=92.42	JasonForJoy (GitHub)
ERMC Sun et al. (2021)	Emotion recog.	MELD; EmoryNLP	F1	Avg F1=64.22	google-research/bert
HeterMPC Gu et al. (2022)	Response gen.	Ubuntu IRC	BLEU; ROUGE	BLEU-1=12.61	lxchtan (GitHub)
SOND Du et al. (2022)	Speaker diarization	AliMeeting	DER	4.46%	yufan-aslp (GitHub)
SDMPED Zhu et al. (2022)	Empathetic gen.	MPED	ROUGE; BLEU	ROUGE-L=12.87	GDPR
PersonaTKG Ju et al. (2022)	(GCN) Persona gen.	HLA-Chat++	PPL; BLEU; Dist	PPL=109.72	NEU-DataMining

Table 4. Cont.

Method	Domain/Task	Datasets	Metrics	Headline Result	Code
User-Aware Park et al. (2022)	Disruptive detect.	ECOJOURNEYS	AUC; PR-AUC	AUC=84.80	gingerit
Hier. VAE Sia et al. (2022)	Argument pred.	CMV	AUC	AUC=69.7	GitHub
E2E Minuting Bhatnagar et al. (2022)	Meeting sum.	XSum; SAMSum	ROUGE; BLEU	R1=45.0; BLEU=7.07	GitHub
PF Li and Zhao (2023)	Response gen.	Ubuntu IRC	BLEU; METEOR	BLEU-1=12.31	EricLee8/MPDRG
ELECTRA-EMVI Li et al. (2023)	Parsing; QA	Molweni	$F1_{RL}$; $F1_G$	$F1_G$ =91.78	EricLee8/MPD_EMVI
MPC-BERT+GIFT Gu et al. (2023)	Addressee; speaker; sel.	Ubuntu IRC	$R_n@1$	$R_2@1$ =95.04	JasonForJoy (GitHub)
MADNet Gu et al. (2023)	MPC generation	Ubuntu IRC	BLEU; METEOR	BLEU-1=11.82	coco-caption
RARM Zhu et al. (2023)	Addressee recog.	Ubuntu IRC	ID/OD-AN	Overall=85.1	None
PFT-Prompt Addlesee et al. (2023)	Goal-tracking; intent-slot	EU SPRING	Accuracy	Acc=62.32/69.57	AddleseeHQ/mpgt-eval
LLM and Multi-Agent Era (2024–2026)					
LLMs zero/few-shot Martinenghi et al. (2024)	Dialogue acts	STAC	Acc; F1	Acc=69.1; F1=71.6	GitHub
Persona-HeterMPC Mahajan and Shaikh (2024)	Persona gen.	Persona-MPC	BLEU; METEOR	BLEU-1=12.47	NEU-DataMining
RL-TRC Fan et al. (2024)	Response gen.	Ubuntu IRC	BLEU; METEOR	BLEU-1=13.66	MaartenGr/KeyBERT
MuPaS Wang et al. (2024)	Generation; speaker pred.	Friends; GoT	Auto + human	GSM8K=43.14	HuggingFace
DDPE Liu et al. (2025)	Discourse parsing	Molweni; STAC	Link; Link+Rel	Link=87.6; L+R=62.9	Shannanliu/DDPE
LIMN Zhou et al. (2025)	Dialogue QA	SQuAD2; Molweni	EM; F1	EM=60.2 (Molweni)	None
CMR Hu et al. (2025)	Response gen.	Friends; Ubuntu IRC	F1; BLEU	F1=13.43	None
RL Fine-tuning Kiruluta et al. (2025)	Multi-turn align.; CoT	ChatEval	Latency	35ms (± 3)	None
DICE-BENCH Jang et al. (2025)	Tool-calling	DICE-BENCH	DICE-Score	3.6444	snuhcc/DICE-Bench
SS-MPC Jang et al. (2025)	Response gen.	Ubuntu IRC	BLEU; ROUGE-L	BLEU-1=15.60	GitHub
BOLT Castillo-López et al. (2025)	Intent recog.	MIntRec2.0; MPGT	Acc; WF1	ACC=41.22/89.47	None
DRCR Cao et al. (2026)	MPD generation	Ubuntu IRC-16/19	BLEU; METEOR	BLEU-1=16.04	None
Context-aware Bhagtani et al. (2026)	Turn-taking	AMI; Friends; SPGI	Acc; F1	61.03/60.54/64.45	GitHub
MPCEval Zhang et al. (2026)	Multi-party eval	DeliData; MPDD	DNR; IR; PF	GPT-4-Turbo results	GitHub
EverMemBench Hu et al. (2026)	Memory eval	EverMemBench	Recall; Acc	37.44/72.61	GitHub
SyntheticMPC Penzo et al. (2026)	Synth. generation	WMPC	Constraints	All=77.72	dhfbk (GitHub)

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